Project Planning Phase
Project Planning

Date	**25 June 2026**
Team ID	
Project Name	**Smart Agricultural Production Optimization Engine (OptiCrop)**
Maximum Marks	5 Marks

Product Backlog, Sprint Planning & Estimation

The Smart Agricultural Production Optimization Engine (OptiCrop) project is planned using the Agile Scrum methodology. The complete project is divided into four development sprints. Each sprint contains one or more epics that are further divided into user stories. Story points are estimated using the Fibonacci sequence (1, 2, 3, and 5) to represent the relative effort required to complete each task.

Product Backlog:

Sprint	Functional Requirement (Epic)	User Story No.	User Story / Task	Story Points	Priority	Team Member(s)
Sprint-1	Dataset Preparation	USN-1	Collect and organize the crop recommendation dataset.	2	High	Sri Teja
Sprint-1		USN-2	Load the dataset into the machine learning environment.	1	High	Sri Teja

Sprint -1	Data Preprocessing	USN-3	Handle missing and inconsistent values.	3	High	Sri Teja
Sprint -1		USN-4	Perform feature preprocessing and prepare training data.	3	High	Sri Teja
Sprint -1		USN-5	Split the dataset into training and testing sets.	2	High	Sri Teja Manikanta
Sprint -2	Machine Learning Model	USN-6	Train Decision Tree model.	2	Medium	Manikanta
Sprint -2		USN-7	Train K-Nearest Neighbour model.	2	Medium	Manikanta
Sprint -2		USN-8	Train Logistic Regression model.	2	Medium	Manikanta
Sprint -2		USN-9	Train Random Forest model.	3	High	Manikanta
Sprint -2	Model Evaluation	USN-10	Compare model performance and select the best model.	3	High	Manikanta
Sprint -2		USN-11	Save the trained model as **model.pkl**.	1	High	Manikanta
Sprint -3	Flask Backend	USN-12	Develop the Flask application structure.	3	High	Manikanta Sri Teja
Sprint -3		USN-13	Implement the prediction module using the trained model.	3	High	Manikanta
Sprint -3		USN-14	Validate user input and handle prediction requests.	2	High	Manikanta Sri Teja
Sprint -3	Frontend Development	USN-15	Design the HTML/CSS user interface.	3	High	Sri Teja

Sprint-3		USN-16	Display crop recommendation results.	2	Medium	Sri Teja
Sprint-4	System Integration	USN-17	Integrate frontend, backend, and machine learning model.	5	High	Manikanta Sri Teja
Sprint-4		USN-18	Perform system testing and validation.	3	High	Manikanta Sri Teja
Sprint-4		USN-19	Resolve defects and improve application stability.	2	Medium	Manikanta Sri Teja
Sprint-4	Deployment	USN-20	Prepare the final application for demonstration and deployment.	2	High	Manikanta Sri Teja

Sprint Tracker :

Sprint	Planned Story Points	Duration	Sprint Start Date	Planned End Date	Story Points Completed	Actual Release Date
Sprint-1	11	1 Day	26/06/26	26/06/26	11	26/06/26
Sprint-2	13	2 Days	26/06/26	28/06/26	13	28/06/26
Sprint-3	13	2 Days	29/06/26	30/06/26	13	01/07/26
Sprint-4	12	1 Day	01/07/26	01/07/26	12	02/07/26

Velocity:

Planned Total Story Points = **49**

Number of Planned Sprints = **4**

Planned Velocity = 49 ÷ 4 ≈ 12 Story Points per Sprint

- **Actual Velocity after Sprint-1:** 11 Story Points
- **Actual Velocity after Sprint-2:** 13 Story Points
- **Actual Velocity after Sprint-3:** 13 Story Points
- **Actual Velocity after Sprint-4:** 12 Story Points
- **Average Actual Team Velocity: 12.25 Story Points per Sprint**

Burndown Chart

Burndown Chart Data:

Timeline Milestone	Date (Actual)	Planned Remaining Points	Actual Remaining Points	Status / Notes
Project Start	26/06/26	49	49	Project kickoff
After Sprint-1	26/06/26	38	38	Completed 11 SP (Dataset & Preprocessing)
After Sprint-2	28/06/26	25	25	Completed 13 SP (ML Modeling)
After Sprint-3	01/07/26	12	12	Completed 13 SP (Backend & Frontend) — *1 day delay*
After Sprint-4	02/07/26	0	0	Completed 12 SP (Integration & Deployment) — *1 day delay*

Burndown Visualization: